Chunking Techniques in RAG Systems

Quick Reference: All Chunking Techniques

The table below summarizes all twelve chunking strategies, their descriptions, ideal use-cases, and compatible file formats.

Chunking Technique	Short Description	Best Used For	Formats
Fixed-size Chunking	Splits text into equal chunks	Any plain text	.txt, .pdf, .docx
Recursive Chunking	Recursively splits by separator priority	Reports, papers	.pdf, .docx, .txt
Header-aware Chunking	Uses headers/sub headers	Markdown docs	.md, .html
Section-aware Chunking	Uses sections/subsections	Research papers	.pdf, .docx
Layout-aware Chunking	Uses visual layout	Resumes, catalogues	.pdf, scanned PDFs
Table-aware Chunking	Preserves tables	Financial reports	.pdf, .xlsx, .csv
Clause-aware Chunking	Preserves legal clauses	Contracts/legal docs	.pdf
OCR-aware Chunking	Handles scanned docs	OCR PDFs, Scanned PDFs	.pdf, .jpg, .png
Function-aware Chunking	Splits code by functions	Source code	.py, .js
Semantic Chunking	Splits by topic changes	Long articles	All text docs
Multi-modal Chunking	Handles text + image + tables	Complex PDFs	.pdf

Chunking Technique	Short Description	Best Used For	Formats
Adaptive Chunking	Dynamically selects strategy	Mixed documents	All formats
Parent-Child Chunking	Stores large parent chunks with smaller child chunks	Long documents, enterprise RAG	.pdf, .docx, .md
DOM-aware Chunking	Uses HTML DOM hierarchy and webpage structure	Web documentation, websites, API docs	.html

1. Research Paper → Section-Aware Chunking

Primary Chunking Technique

Section-Aware Chunking
A text-splitting strategy that uses a document's actual layout and formatting — such as headers, sections, and chapters — rather than just punctuation or character counts to divide text.

How It Works

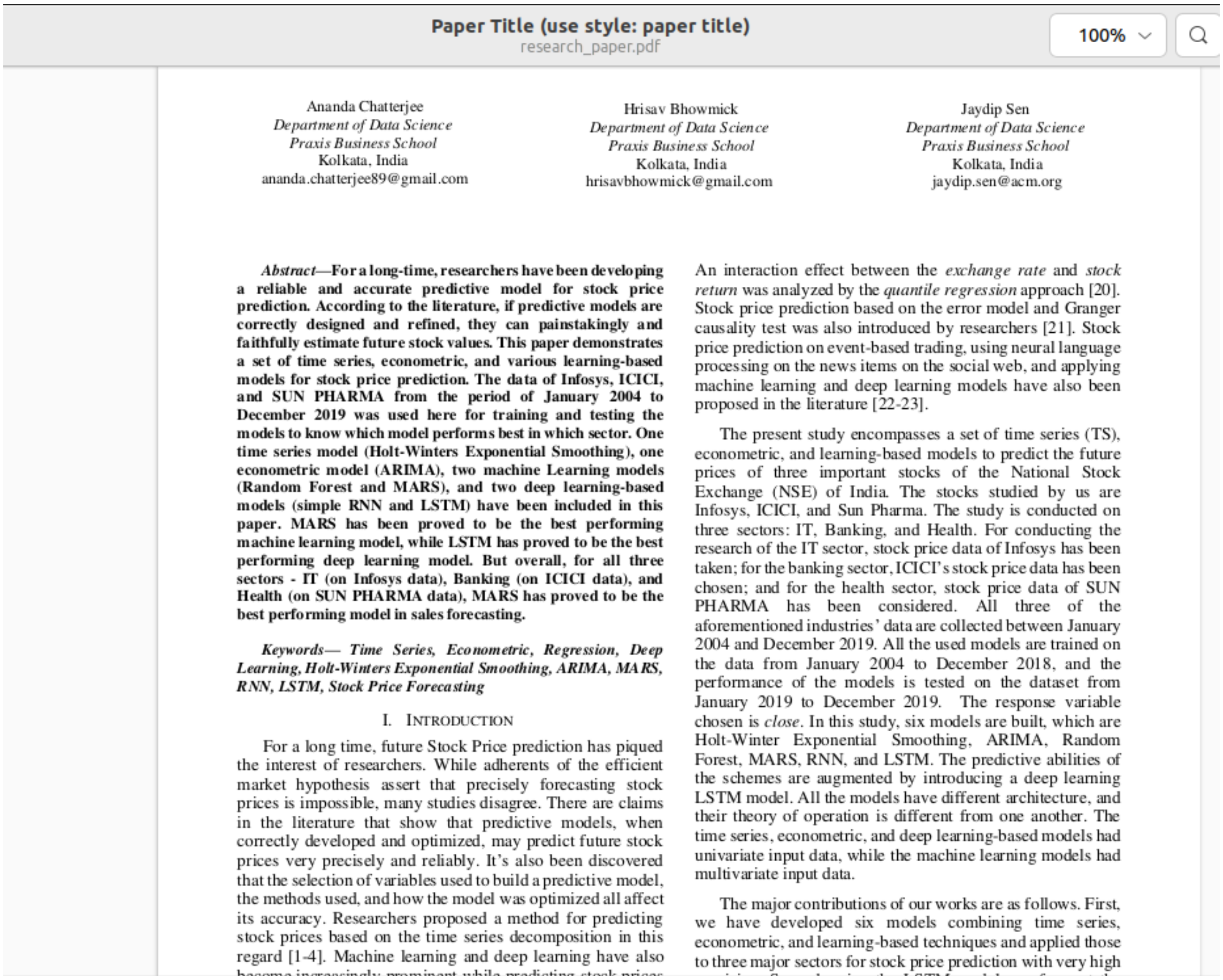
Instead of guessing where a topic ends based on paragraphs, this method actively parses structural tags like <h1>, <h2>, #, or Markdown headers. It groups all text under a specific heading into a single chunk, ensuring that everything discussing the same sub-topic stays together.

Other Possible Techniques

Technique	Reason
Recursive	Paragraph preservation
Semantic	Topic continuity
Layout-aware	Multi-column handling

Why Used

The document contains semantic sections such as Abstract, Introduction, Problem Statement, and Methodology — making structural splitting the most logical approach.



Paper Title (use style: paper title)
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***Abstract*—For a long-time, researchers have been developing a reliable and accurate predictive model for stock price prediction. According to the literature, if predictive models are correctly designed and refined, they can painstakingly and faithfully estimate future stock values. This paper demonstrates a set of time series, econometric, and various learning-based models for stock price prediction. The data of Infosys, ICICI, and SUN PHARMA from the period of January 2004 to December 2019 was used here for training and testing the models to know which model performs best in which sector. One time series model (Holt-Winters Exponential Smoothing), one econometric model (ARIMA), two machine Learning models (Random Forest and MARS), and two deep learning-based models (simple RNN and LSTM) have been included in this paper. MARS has been proved to be the best performing machine learning model, while LSTM has proved to be the best performing deep learning model. But overall, for all three sectors - IT (on Infosys data), Banking (on ICICI data), and Health (on SUN PHARMA data), MARS has proved to be the best performing model in sales forecasting.**

Keywords— Time Series, Econometric, Regression, Deep Learning, Holt-Winters Exponential Smoothing, ARIMA, MARS, RNN, LSTM, Stock Price Forecasting

I. INTRODUCTION

For a long time, future Stock Price prediction has piqued the interest of researchers. While adherents of the efficient market hypothesis assert that precisely forecasting stock prices is impossible, many studies disagree. There are claims in the literature that show that predictive models, when correctly developed and optimized, may predict future stock prices very precisely and reliably. It's also been discovered that the selection of variables used to build a predictive model, the methods used, and how the model was optimized all affect its accuracy. Researchers proposed a method for predicting stock prices based on the time series decomposition in this regard [1-4]. Machine learning and deep learning have also

An interaction effect between the *exchange rate* and *stock return* was analyzed by the *quantile regression* approach [20]. Stock price prediction based on the error model and Granger causality test was also introduced by researchers [21]. Stock price prediction on event-based trading, using neural language processing on the news items on the social web, and applying machine learning and deep learning models have also been proposed in the literature [22-23].

The present study encompasses a set of time series (TS), econometric, and learning-based models to predict the future prices of three important stocks of the National Stock Exchange (NSE) of India. The stocks studied by us are Infosys, ICICI, and Sun Pharma. The study is conducted on three sectors: IT, Banking, and Health. For conducting the research of the IT sector, stock price data of Infosys has been taken; for the banking sector, ICICI's stock price data has been chosen; and for the health sector, stock price data of SUN PHARMA has been considered. All three of the aforementioned industries' data are collected between January 2004 and December 2019. All the used models are trained on the data from January 2004 to December 2018, and the performance of the models is tested on the dataset from January 2019 to December 2019. The response variable chosen is *close*. In this study, six models are built, which are Holt-Winter Exponential Smoothing, ARIMA, Random Forest, MARS, RNN, and LSTM. The predictive abilities of the schemes are augmented by introducing a deep learning LSTM model. All the models have different architecture, and their theory of operation is different from one another. The time series, econometric, and deep learning-based models had univariate input data, while the machine learning models had multivariate input data.

The major contributions of our works are as follows. First, we have developed six models combining time series, econometric, and learning-based techniques and applied those to three major sectors for stock price prediction with very high

Example Chunks

- Chunk 1 — Section: **Abstract** For a long time, researchers have been developing a reliable and accurate predictive…
- Chunk 2 — Section: **Introduction** For a long time, future stock price prediction…

Technique 2: Parent–Child Chunking

The document contains major sections, subsections, and long contextual content that benefit from a hierarchical chunking approach:

PDF

↓

Section Split

↓

Parent Chunks

↓

Smaller Child Chunks

↓

Embed ONLY Child Chunks

↓

Retrieve Child

↓

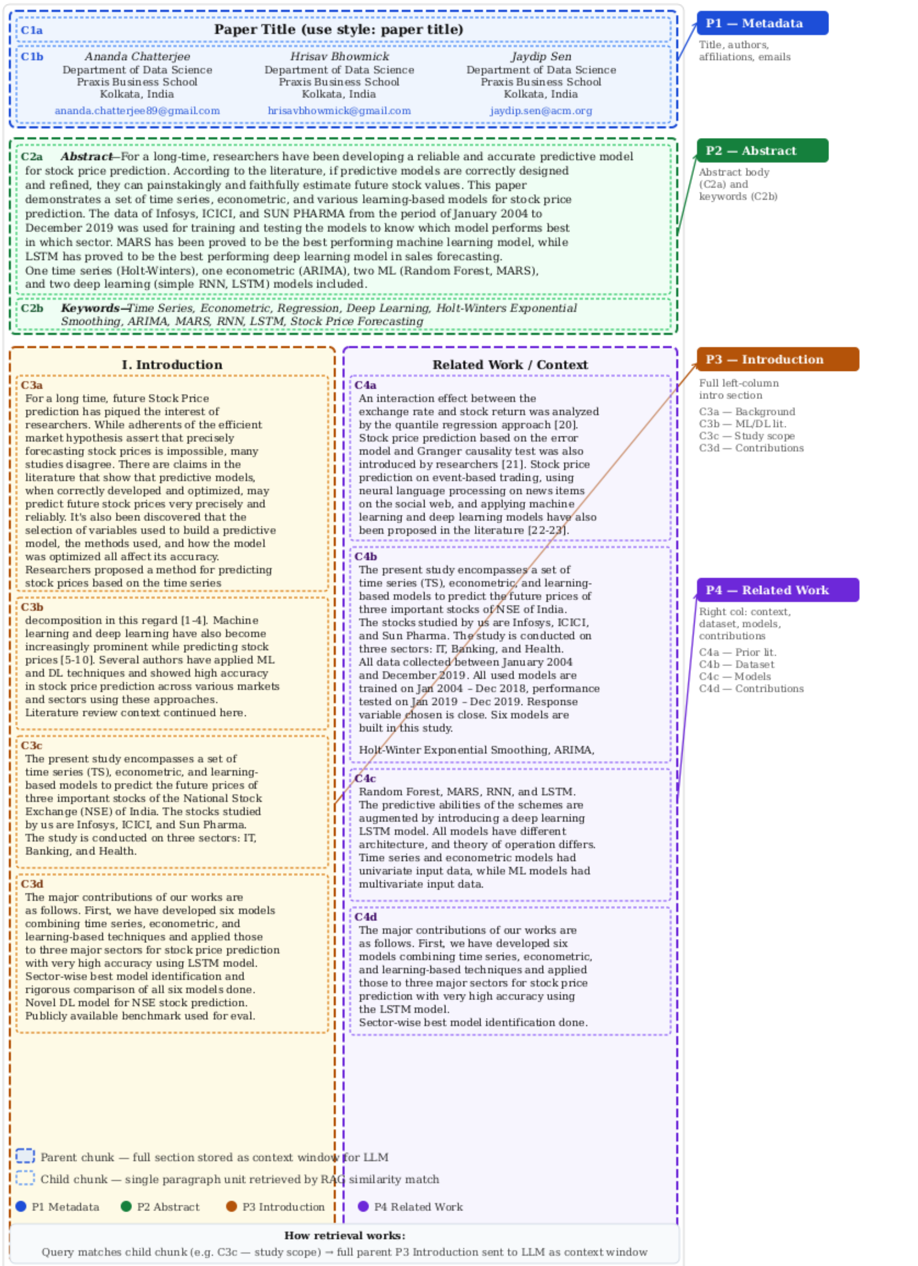
Return Parent Context

Parent Chunk Large semantic section (keeps full context)	**Child Chunk** Small retrievable chunks (created from parent)

The system retrieves small child chunks and then returns the related parent chunk for better context.

Example Chunks

- P --> Parent Chunks
- C --> Child Chunks



2. Resume → Layout-Aware Chunking

Primary Chunking Technique

Layout-Aware Chunking
An advanced text-splitting strategy that uses a document's physical, visual layout — like tables, columns, headers, footers, and sidebars — to divide text into logical chunks. While section-aware chunking looks at text formatting tags, layout-aware chunking uses a document's spatial design to understand context.

Why Used

The document contains visually separated sections such as Education, Work Experience, and Technical Skills that are distinguished by spatial design rather than semantic headers.

UNIVERSITY OF CENTRAL MISSOURI
CAREER AND LIFE DESIGN

FIRST NAME LAST NAME

(XXX) XXX-XXXX | ProfessionalEmail@gmail.com | linkedin.com/in | City, State

EDUCATION

Degree, Major (e.g Bachelor of Science, Communication Studies) Graduating Month Year
Name of University, Institution City, State
Minor: XXXXX GPA x.x/4.0 (optional but recommend including if over a 3.0, can be Overall GPA or Major GPA)

If you have a Masters Degree then repeat above format for undergraduate with your highest level of education appearing first
Do not use bullet points in the education section and do not include high school information

WORK EXPERIENCE (could also be titled Relevant Experience or Related Experience)

Position Title ***Roles are in reverse chronological order with most recent first*** Month Year - Month Year
Organization/Company Name City, State

- Write your main highlighted accomplishments.
- Think about how your task/project helped the company do better and how you added value to the company.
- Follow the format “Performed X by doing Y resulting in Z”, quantify with numbers, percentages, other data where you can
- Start with strong action verbs and avoid using responsible, helped, researched or assisted (review our Guide to Using Action Verbs).

Position Title Month Year - Month Year
Organization Name City, State

- Highlight different skills
- Show overall benefit to the organization so that future employers know how you could add value to their organization
- You resume should demonstrate competencies: Critical thinking, leadership, teamwork, communication, career management etc.
- Don't forget to include essential (soft) skills and technical (hard) skills where relevant.

Position Title Month Year - Month Year
Organization Name City, State

- Roles with more relevant experiences to the job you are applying for should have more bullet points (no more than 5)
- Roles with less relevant experiences should have less bullet points especially if you're running out of space
- E.g.: For working as a lifeguard: “Supervised over 100 kids during the summer achieving 0 incidents and 97% open rate for the local community pool, during business hours.”

SECTION HEADER

Title Month Year - Month Year

Example Chunks

- Chunk 1 — Section: Education   Degree, Major | University Name | GPA
- Chunk 2 — Section: Work Experience   Position Title | Organization Name | Accomplishments

3. Legal Document → Clause-Aware Chunking

Primary Chunking Technique

Clause-Aware Chunking
A granular text-splitting strategy used primarily for legal documents, contracts, and regulatory compliance. It breaks down text based on legal clauses, sub-clauses, or specific provisions (like "Section 4.1(a)") rather than relying on standard sentence boundaries or character counts.

How It Works

This method leverages specialized parsers, Regular Expressions (RegEx), or NLP models trained on legal syntax to recognize the unique numbering and phrasing of legal documents:

- Identification: Scans for legal indicators such as "Provided that," "Notwithstanding," "In witness whereof," or alphanumeric numbering systems (e.g., 1.1, 1.1.1, (a), (i)).
- Isolation: Treats each individual clause as a distinct unit of meaning, ensuring that a condition, obligation, or right is never separated from its subject.
- Hierarchy Mapping: Appends parent context to sub-clauses. For instance, if sub-clause (ii) is nested under Section 4 (Indemnification), the chunker passes the context of Section 4 into the chunk for (ii).

Why Used

Legal meaning depends on preserving clauses, agreements, and legal conditions together to avoid misinterpretation.

GENERAL SALE DEED

This deed of sale is made and executed on this .... day of .... 20...

By

Sri/Smt................ S/o/W/o..................,Occupation................Aged....................years, residing at .... (Principal), represented by his agent Sri... S/o........................... ,...................... Occupation .... aged............ Years, residing at .... ,..........................by means of a General/Special Power of Attorney dated ,............................registered/authenticated as Document No............................. of Book IV of Sub Registrar's Office... hereinafter called the

"VENDOR"

(which expression shall wherever it occurs in this deed includes the said person, his/her heirs, legal representatives, agents, executors, administrators, assignees or any person claiming through or under him) of one part.

In favour of

Sri/Smt..............................S/o.D/o..............................Occupation.....,........................ ....aged............ years, residing at .... hereinafter called the

"VENDEE"

(which expression shall wherever it occurs in this deed includes the said person, his/her heirs, legal representatives, agents, executors, administrators, assignees or any person claiming through or under him) of other part.

Whereas the Vendor is the absolute owner, having acquired the property, which is more specifically and clearly delineated in the schedule hereto, by inheritance / by partition of joint family properties/ by release/by gift / by gift settlement / by will / by sale executed by.......................................and registered as document No............of................... Of Book.....................Volume No.......................Pages..................in the office of the Registrar / Sub-Registrar.. and since then he is in the possession and absolute enjoyment thereof.

And

Whereas the vendor intends to sell away the said property wherein he or she has got good and marketable title, rights, interest and possession and no other person has got any right, title or interest over the schedule property.

And

Whereas the Vendee offered to purchase it for a sum of Rs.......................for which the Vendor accepted and has agreed to sell the same to the Vendee.

Example Chunks

- Chunk 1 — Metadata / Parties GENERAL SALE DEED — This deed of sale is made and executed on this … day … By: Vendor details … In favour of:
- Chunk 2 — Ownership Background Whereas the Vendor is the absolute owner … acquired the property by inheritance / partition / gift / sale…
- Chunk 3 — Intent to Sell Whereas the vendor intends to sell away the said property … Whereas the Vendee offered to purchase…

4. README / Markdown → Header-Aware Chunking

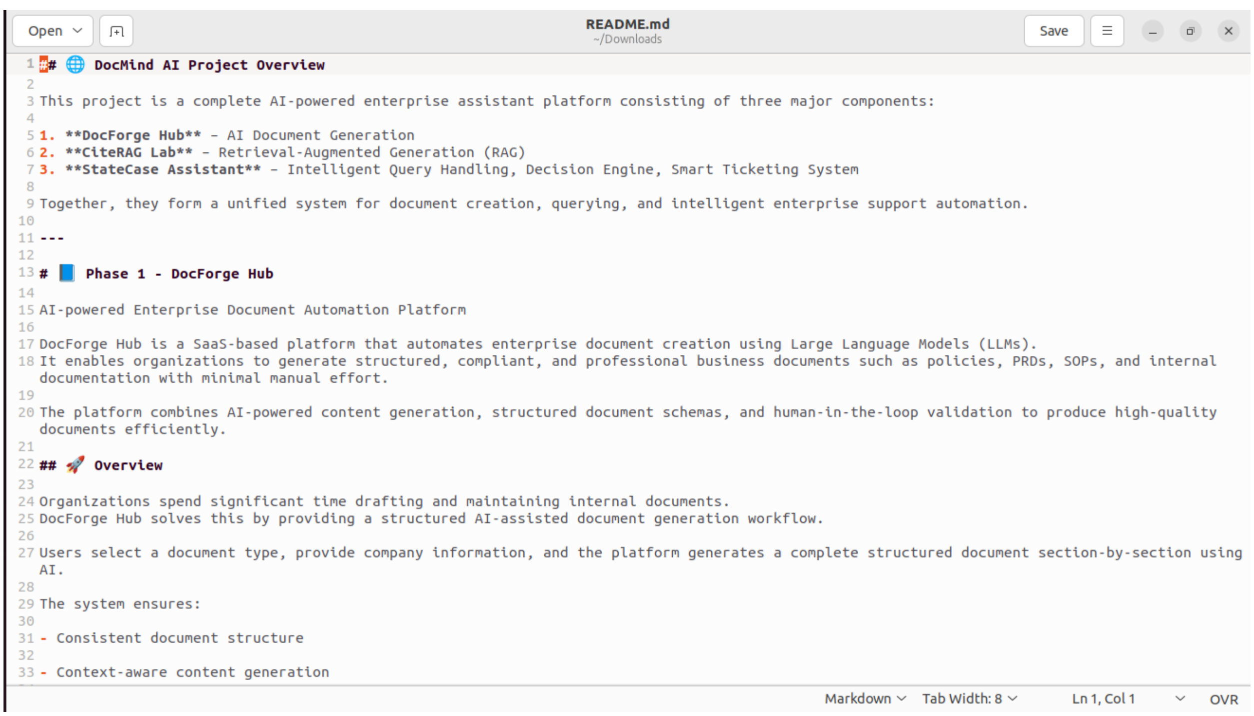
Primary Chunking Technique

> **Header-Aware Chunking**
> A text-splitting strategy that uses a document's heading hierarchy (like #, ##, ### in Markdown or <h1>, <h2> in HTML) to determine where to split text. It ensures that content nested under a specific title stays together alongside its heading.

How It Works

Instead of splitting text purely by character count, the chunker actively looks for header tags:

- Header Detection: The algorithm scans the document for structural headers.
- Grouping: Bundles all paragraphs, bullet points, and text that fall under a specific header into a single chunk.
- Context Injection: When a chunk is created, it appends the higher-level headers as metadata. For example, text under ### Features will also carry the context of its parent # Product Guide.



```
# 🌐 DocMind AI Project Overview

This project is a complete AI-powered enterprise assistant platform consisting of three major components:

1. **DocForge Hub** – AI Document Generation
2. **CiteRAG Lab** – Retrieval-Augmented Generation (RAG)
3. **StateCase Assistant** – Intelligent Query Handling, Decision Engine, Smart Ticketing System

Together, they form a unified system for document creation, querying, and intelligent enterprise support automation.

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# 📘 Phase 1 - DocForge Hub

AI-powered Enterprise Document Automation Platform

DocForge Hub is a SaaS-based platform that automates enterprise document creation using Large Language Models (LLMs).
It enables organizations to generate structured, compliant, and professional business documents such as policies, PRDs, SOPs, and internal documentation with minimal manual effort.

The platform combines AI-powered content generation, structured document schemas, and human-in-the-loop validation to produce high-quality documents efficiently.

## 🚀 Overview

Organizations spend significant time drafting and maintaining internal documents.
DocForge Hub solves this by providing a structured AI-assisted document generation workflow.

Users select a document type, provide company information, and the platform generates a complete structured document section-by-section using AI.

The system ensures:

- Consistent document structure

- Context-aware content generation
```


Example Chunks

- Chunk 1 — Header: DocMind AI Project Overview

- Chunk 2 — Header: Phase 1 – DocForge Hub

5. Financial Report → Table-Aware + Section-Aware Chunking

Primary Chunking Technique

> **Table-Aware + Section-Aware Chunking**
> A hybrid text-splitting strategy that combines document structure with tabular data recognition. It ensures that a document's hierarchical narrative (sections) and its structured data sets (tables) are parsed without losing their meaning or formatting.

How It Works

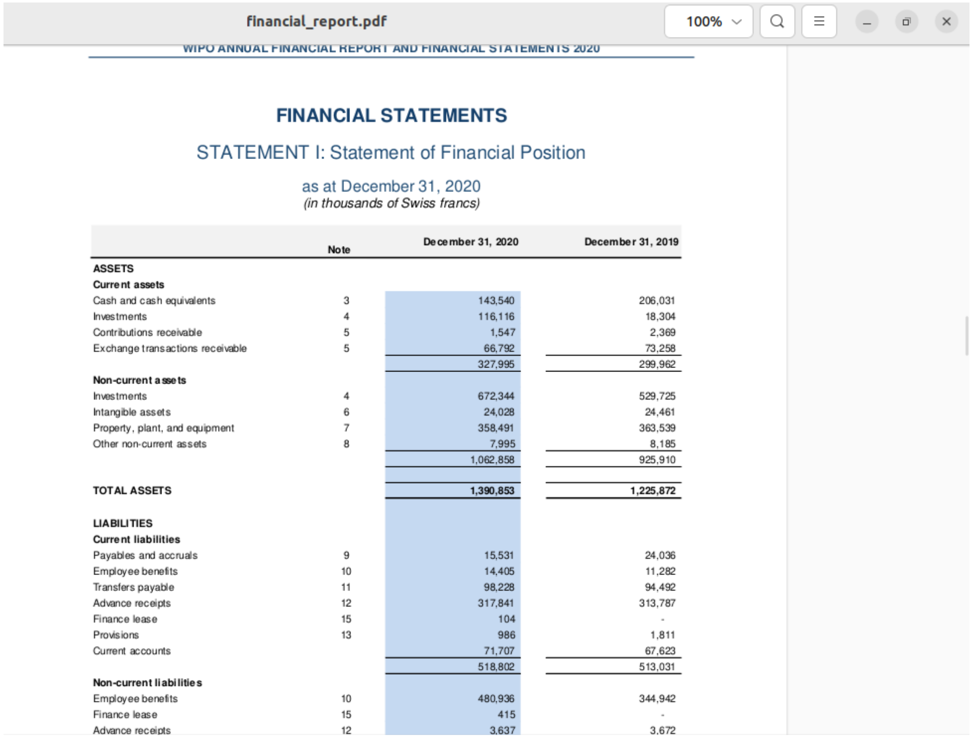
This dual-layered approach processes a document by running two distinct logic tracks simultaneously:

Layer 1: Section-Aware (The Narrative)

- Scans the document for structural markers like headers (#, ##, <h1>), chapters, or explicit section breaks.
- Groups paragraphs, lists, and sub-headings belonging to a specific section into a unified chunk.

Layer 2: Table-Aware (The Data)

- Standard chunkers read tables row-by-row as plain text, scrambling columns. This layer isolates the table and detects its boundaries.
- Preserves table structure by converting to Markdown tables, HTML <table> tags, or JSON.
- Ensures the entire table (or logical sub-sections of a large table) is kept as a single, unbreakable chunk.


financial_report.pdf

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WIPO ANNUAL FINANCIAL REPORT AND FINANCIAL STATEMENTS 2020

FINANCIAL STATEMENTS

STATEMENT I: Statement of Financial Position

as at December 31, 2020

(in thousands of Swiss francs)

	Note	December 31, 2020	December 31, 2019
ASSETS			
Current assets			
Cash and cash equivalents	3	143,540	206,031
Investments	4	116,116	18,304
Contributions receivable	5	1,547	2,369
Exchange transactions receivable	5	66,792	73,258
		327,995	299,962
Non-current assets			
Investments	4	672,344	529,725
Intangible assets	6	24,028	24,461
Property, plant, and equipment	7	358,491	363,539
Other non-current assets	8	7,995	8,185
		1,062,858	925,910
TOTAL ASSETS		**1,390,853**	**1,225,872**
LIABILITIES			
Current liabilities			
Payables and accruals	9	15,531	24,036
Employee benefits	10	14,405	11,282
Transfers payable	11	98,228	94,492
Advance receipts	12	317,841	313,787
Finance lease	15	104	-
Provisions	13	986	1,811
Current accounts		71,707	67,623
		518,802	513,031
Non-current liabilities			
Employee benefits	10	480,936	344,942
Finance lease	15	415	-
Advance receipts	12	3,637	3,672

Example Chunks

- Chunk 1 — Section: Statement of Financial Position
- Chunk 2 — Table: Cash Flow 2020

6. Code → Function-Aware Chunking

Primary Chunking Technique

Function-Aware Chunking

Function-aware chunking (often referred to as Code-aware chunking) is a specialized structural text-splitting strategy designed specifically for source code. Instead of cutting code by lines or character limits—which breaks syntax and destroys programming logic—it parses the source code into its abstract syntax tree (AST) and splits it based on complete logical blocks like **functions, classes, or methods**.

How It Works

Code Upload
↓
Programming Language Detection
↓
Parser / AST Generation
↓
Detect:
- functions
- classes
- methods
- imports
↓
Create Function Chunks
↓
Optional Parent-Child Chunking
↓
Generate Embeddings
↓
Store in Vector DB
↓
Semantic Code Retrieval

```python
import pandas as pd
from sklearn.linear_model import LinearRegression

def load_data(path):
    df = pd.read_csv(path)
    return df

def preprocess(df):
    df = df.dropna()
    df["price"] = df["price"] / 100
    return df

def train_model(df):
    X = df[["feature1", "feature2"]]
    y = df["price"]

    model = LinearRegression()
    model.fit(X, y)

    return model

def predict(model, data):
    return model.predict(data)
```

Example Chunk

- **Chunk 1 – Imports**

  import pandas as pd

  from sklearn.linear_model import LinearRegression

- **Chunk 2 – load_data()**

  def load_data(path):

      df = pd.read_csv(path)

      return df

- **Chunk 3 - Preprocess ()**

  def preprocess(df):

      df = df.dropna()

      df["price"] = df["price"] / 100

      return df

7. Scanned PDF → OCR-Aware Chunking

Primary Chunking Technique

OCR-Aware Chunking
A text-splitting strategy designed specifically for scanned documents, images, and unsearchable PDFs. It integrates Optical Character Recognition (OCR) directly into the chunking pipeline, translating physical pixels and visual layouts into structured, contextual text chunks.

Standard chunkers fail on scanned documents because they see a page as a flat string of poorly extracted text (often riddled with layout errors). OCR-aware chunking solves this by utilizing spatial and visual coordinates.

How It Works

- Bounding Box Detection: The OCR engine identifies words, lines, and blocks of text, assigning each a set of X, Y coordinates on the page.
- Visual Block Segmentation: The chunker uses coordinates to group text blocks that belong together visually (e.g., a paragraph block, a sidebar, or a distinct column).
- Reading Order Sorting: Reconstructs the natural reading order. For scanned pages with two columns, ensures column one is fully read before column two.
- Chunking with Spatial Metadata: Each chunk is tagged with its source page number and visual coordinates (X, Y bounding box).

Sample Scanned Copy:

Academic Year : 2012-2013

ANNA UNIVERSITY, CHENNAI
FACULTY OF SCIENCE AND HUMANITIES
CHENNAI - 600 025. INDIA

Prof. K. SHANTHI, Ph.D
Chairperson

Phone : +91-44-22359488
+91-44-22358654
Cell : +91-9840140642
+91-9962520678
shanthiramesh@annauniv.edu

24.05.2013

<u>To whom so ever it may concern</u>

This is to certify **that Dr. O.D.Shakila. Associate Professor of Chemistry, K.L.N. College of Engineering, Pottapalayam- 630611** has attended the Anna University – 18th meeting of Board of Studies at Anna University on 24.05.2013.

24/05/2013.

(Dr.K.SHANTHI)

(Chairman, Faculty of S&H)

Example Chunk

- Chunk 1 — ANNA UNIVERSITY, CHENNAI FACULTY OF SCIENCE AND HUMANITIES Chennai – 600 025, INDIA Prof. K. SHANTHI, Ph.D. Chairperson Phone: +91-44-22359488 Cell: +91-9840149642 / +91-9962520678 shanthiramesh@annauniv.edu
- Chunk 2 — 24.05.2013 To whom so ever it may concern

- Chunk 3 — This is to certify that Dr. O.D.Shakila, Associate Professor of Chemistry, K.L.N. College of Engineering, Pottapalayam- 630611 has attended the Anna University – 18th meeting of Board of Studies at Anna University on 24.05.2013.

- Chunk 4 — (Dr. K.SHANTHI) (Chairman, Faculty of S&H) [handwritten date: 24/05/2013]

Adaptive Chunking

Overview

Adaptive Chunking is an advanced technique used in RAG (Retrieval-Augmented Generation) systems to split documents intelligently instead of using fixed-size chunks. In traditional chunking, documents are divided after a fixed number of words or tokens — simple but often breaks meaning and structure.

Adaptive Chunking solves this by understanding the structure and type of the document before splitting it. It dynamically changes the chunking strategy depending on the content:

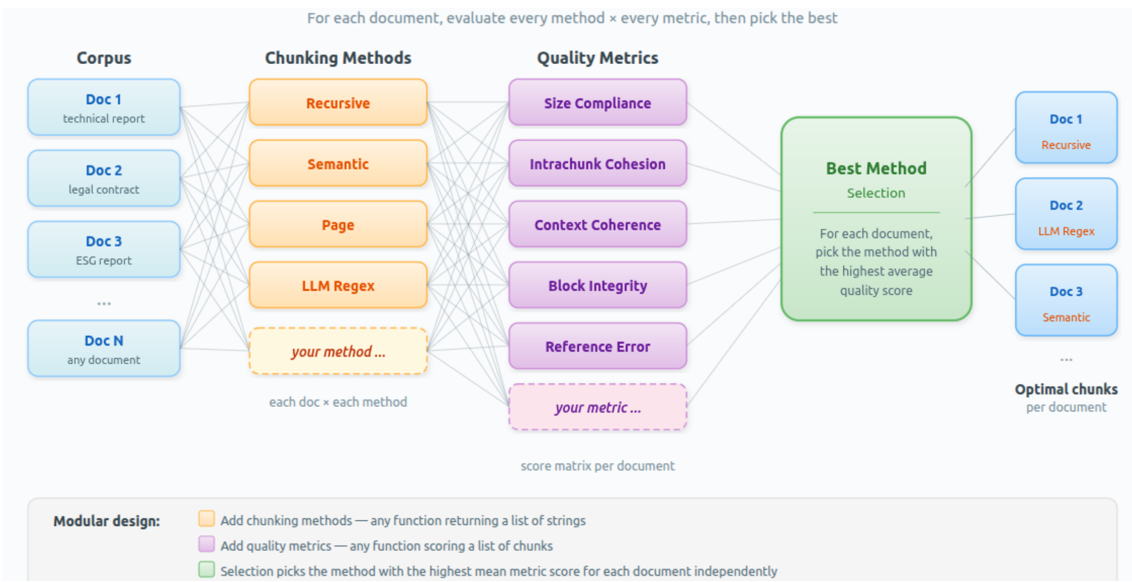
- Research papers: chunks created using sections — Introduction, Methodology, Conclusion.
- Code files: chunks created function-wise or class-wise.
- Legal documents: chunks divided clause-wise or article-wise.
- FAQ documents: each question-answer pair becomes a chunk.

This method preserves semantic meaning and keeps related information together. Adaptive Chunking may use:

- Semantic analysis
- Document hierarchy
- Headings and formatting
- Layout information
- Topic changes
- Code structure
- Tables and images

Advantages of Adaptive Chunking

- Preserves document structure
- Improves semantic understanding
- Gives more accurate retrieval
- Works well for complex documents
- Handles different document types intelligently
- Reduces irrelevant context retrieval



Adaptive Chunking — Pipeline Flows

General Adaptive Chunking Pipeline

Document Upload
↓
Document Type Detection
↓
Structure / Layout Analysis
↓
Select Chunking Strategy
↓
Adaptive Chunk Creation
↓
Embedding Generation
↓
Vector Database Storage
↓
Retrieval

1. Research Paper Flow

Research Paper Upload
↓
PDF Parsing
↓
Detect Research Paper Structure
↓
(Abstract, Introduction, References)
↓
Layout & Heading Analysis
↓
Section-Aware Chunking
↓
Semantic Chunk Refinement
↓

Table / Figure Handling

↓

Parent-Child Chunk Creation

↓

Embedding Generation

↓

Vector Database Storage

↓

Semantic Retrieval

2. Code File Flow

Code File Upload
↓
Language Detection
↓
AST Parsing (Abstract Syntax Tree)
↓
Detect: Classes, Functions, Imports, Methods
↓
Function-Aware Chunking
↓
Dependency Mapping
↓
Parent-Child Chunking (Large Functions)
↓
Code Embedding Generation
↓
Vector Database Storage
↓
Code Retrieval

3. Legal Document Flow

Legal Document Upload
↓
OCR (if scanned)
↓
Legal Structure Detection
↓
Detect: Clauses, Articles, Sections, Subsections
↓
Clause-Aware Chunking
↓
Cross-Reference Preservation
↓
Legal Metadata Extraction
↓

Embedding Generation
↓
Vector Database Storage
↓
Clause-Level Retrieval

4. Financial Report Flow

Financial Report Upload

↓

Document Classification

↓

Table Detection + Layout Analysis

↓

Detect: Balance Sheets, Revenue Tables, Risk Sections, Cash Flow

↓

Table-Aware Chunking

↓

Section-Aware Chunking

↓

Numeric Context Preservation

↓

Embedding Generation

↓

Vector Database Storage

↓

Financial Semantic Retrieval

5. Scanned PDF Flow

Scanned PDF Upload

↓

OCR Processing (PaddleOCR / Tesseract / Surya)

↓

Layout Detection

↓

Text Reconstruction

↓

Adaptive Chunking

↓

Embedding Generation

↓

Vector Database Storage

↓

OCR-Based Retrieval

Multi-modal Chunking

Primary Chunking Technique

> **Multi-modal chunking**
>
> **Multi-modal chunking** is the process of splitting a document that contains multiple data formats—such as plain text, structural tables, and embedded images—into unified, information-rich pieces.
>
> Instead of isolating or throwing away non-text assets, multimodal chunking processes diverse content types simultaneously, preserving the vital spatial and semantic connections between what you read and what you see.

Workflow

Document Upload

↓

OCR + Layout Analysis

↓

Detect Modalities

(text/image/table/chart)

↓

Separate Content Blocks

↓

Link Related Modalities

↓

Multimodal Chunk Creation

↓

Text + Vision Embeddings

↓

Vector Database Storage

↓

Multimodal Retrieval

DOM – Aware Chunking (Document Object Model)

Primary Chunking Technique

DOM Aware Chunking

DOM-aware chunking is a structural text-splitting strategy designed specifically for web pages, HTML documents, and XML files.

Instead of relying on basic text rules like paragraph breaks or character counts, this method relies on the **Document Object Model (DOM)** tree structure—the underlying nested tag code that builds every website—to divide content into logical, cohesive pieces.

Workflow

HTML/Webpage

↓

HTML Parser

↓

DOM Tree Construction

↓

Tag & Structure Detection

↓

Section Identification

↓

Chunk Boundary Detection

↓

Chunk Generation

↓

Metadata Attachment

↓

Embeddings & Vector DB

Original HTML

```
<section>
  <h1>API Authentication</h1>

  <p>JWT token generation process.</p>

  <code>
    const token = jwt.sign(...)
  </code>
</section>
```

DOM-aware Chunks

Chunk 1 — Text Chunk

Heading: API Authentication

JWT token generation process.

Chunk 2 — Code Chunk

Code Block:
const token = jwt.sign(...)